USN. | | | | | | | | | | [22MTEST312]

THIRD SEMESTER M.Tech EXAMINATION, DECEMBER-2024

EMBEDDED SYSTEMS FOR SMART APPLIANCES AND ENERGY MANAGEMENT

Time: 3 Hours **Maximum Marks: 100**

Instructions:
- **i.** Answer **FIVE** full Questions. Question number 1 and 2 are compulsory. Each question carries 20 marks.
- **ii.** Missing data may be suitably assumed.

1.	**a.**	Explain how the principles and challenges of energy production influence the functioning of the energy market.	**CO1**	**L2**	**10**
	b.	Design a comprehensive Demand-Response Management (DRM) strategy for a smart grid system using SmartCoDe's classification of energy.	**CO1**	**L3**	**10**
2.	**a.**	Discuss the economic pressures driving change in energy systems, focusing on the Integrated Distributed Energy Systems (IDES) paradigm.	**CO2**	**L2**	**10**
	b.	Discuss how innovations in energy policy and legislation can support the development of the Future.	**CO2**	**L3**	**10**
3.	**a.**	Analyze the importance of wireless sensor network (WSN) modeling in improving the efficiency of smart appliances in a smart home or building.	**CO3**	**L3**	**10**
	b.	Explain the differences between TLM and WSN models in terms of transaction management and their applications in system design.	**CO3**	**L3**	**10**
		OR			
4.	**a.**	Explain the role of Electronic System Level (ESL) simulation in the SmartCoDe project.	**CO3**	**L2**	**10**
	b.	Explain the process of abstract modeling in the design of smart appliances with an example.	**CO3**	**L2**	**10**
5.	**a.**	Explain how ZigBee facilitates energy-efficient communication in building automation systems and why it is a preferred choice for smart appliances.	**CO4**	**L2**	**10**
	b.	Compare the strengths and weaknesses of Wireless HART and ISA100 protocols in industrial building automation, focusing on reliability, scalability, and data security.	**CO4**	**L2**	**10**
		OR			

JAIN (Deemed-to-be University) Learning Resource Centre Jain Global Campus Ramanagar-112

6.	**a.**	Compare the physical layer properties, extra features, and limitations of different wireless sensor network protocols.	**CO4**	**L2**	**10**
	b.	Discuss the maximum number of devices supported and the signal ranges of various wireless network protocols.	**CO4**	**L2**	**10**
7.	**a.**	Discuss the key principles involved in designing secure systems for smart, embedded environments.	**CO5**	**L2**	**10**
	b.	Explain the current challenges in the secure key distribution protocols for smart embedded appliances networks.	**CO5**	**L2**	**10**
		OR			
8.	**a.**	Discuss strategies to mitigate the potential security threats to wireless communication networks in smart embedded appliances.	**CO5**	**L2**	**10**
	b.	Explain the following terms related to smart embedded application networks and their significance in the context of security within building automation systems: (a) Wireless HART, (b) End-to-End Encryption, and (c) Man-in-the-Middle Attack.	**CO5**	**L2**	**10**